Examiner only

4. An investigation compared the composition of inspired and expired air. This is shown in the table below.

Gas	% Concentration of air	
	inspired	expired
oxygen	20.9	16.0
carbon dioxide	0.04	4.0
water vapour	variable	variable
nitrogen	78.1	78.1

(a) (i) Calculate the difference in the % concentration of oxygen in the expired air and the inspired air. [1]

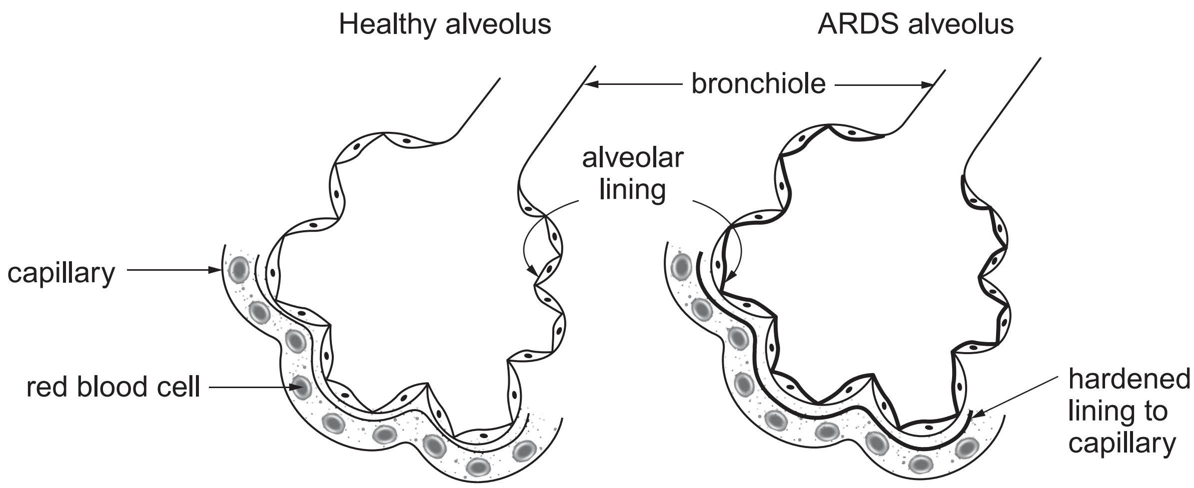
difference = ... %

(ii) State the process in cells that uses oxygen and glucose to release energy. [1]

....

(b) People with a disease called ARDS (Acute Respiratory Distress Syndrome) have difficulty getting enough oxygen.

The diagrams show a healthy alveolus and an alveolus from someone with ARDS.



(i) **Draw an arrow on the diagram** of the **healthy** alveolus, to show the direction of movement of oxygen through the alveolar lining. [1]

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(ii) Describe **two** differences you can see between the two diagrams and explain why people with ARDS have difficulty getting enough oxygen from inspired air. [3]

....

....

....

....

....

(c) Many different types of molecules pass across cell membranes.

Complete the table below to show the direction of movement of molecules between blood and muscles. Place **one** tick (✓) in each row. [3]

Molecule	**From** blood **to** muscles	**From** muscles **to** blood	**To and from** blood and muscles
oxygen			
carbon dioxide			
water			

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